



DATA ANALYSIS OF LOKSABHA GENERAL ELECTIONS/BY-ELECTIONS AT CANDIDATE LEVEL

Abstract

This Excel project analyzes the Lok Sabha General Elections/By-Election at Candidate Level dataset to examine candidate performance, party trends, voter turnout, and election outcomes across India. Using data cleaning, transformation, and data modeling techniques, the project leverages Excel's Power Pivot and Pivot Tables to generate meaningful electoral insights and support data-driven analysis.

Infanta Kennedy
Entri

Data pre-processing using Excel

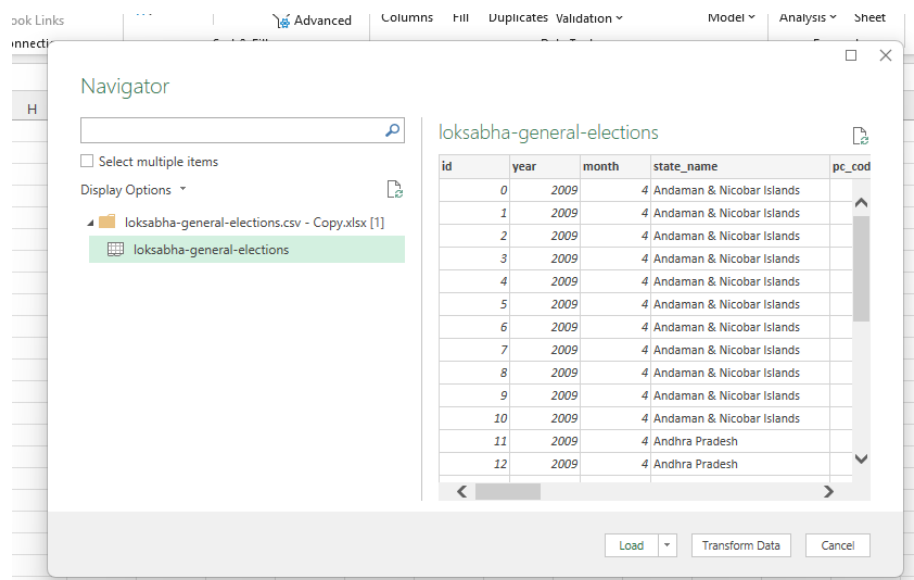
Loading of election data set into the excel:

I have chosen Loksabha general elections/by-elections at candidate level details dataset here for reviewing and applying excel applications. In order to import a excel dataset, I have first checked the type of excel file by clicking their property and load the data set by providing data->get data->from file->from excel_workbook

Dataset before Import:

P13

Dataset after Import:



id	year	month	state_name	pc_code
0	2009	4	Andaman & Nicobar Islands	
1	2009	4	Andaman & Nicobar Islands	
2	2009	4	Andaman & Nicobar Islands	
3	2009	4	Andaman & Nicobar Islands	
4	2009	4	Andaman & Nicobar Islands	
5	2009	4	Andaman & Nicobar Islands	
6	2009	4	Andaman & Nicobar Islands	
7	2009	4	Andaman & Nicobar Islands	
8	2009	4	Andaman & Nicobar Islands	
9	2009	4	Andaman & Nicobar Islands	
10	2009	4	Andaman & Nicobar Islands	
11	2009	4	Andhra Pradesh	
12	2009	4	Andhra Pradesh	

	A	B	C	D	E	F	G	
1	id	year	month	state_name	pc_code	pc_name	pc_type	candidate_name
2	0	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Bishnu P.
3	1	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Kuldeep F
4	2	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri Tapan Kur
5	3	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. P. R. Gan
6	4	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Smti. R. S. Um
7	5	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Pradeep I
8	6	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Vakiath V
9	7	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. M. S. Moh
10	8	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. N. K. P. N
11	9	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Dr. Thankacha
12	10	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. T. Ali
13	11	2009	4	Andhra Pradesh	1	Adilabad	ST	Rathod Rames
14	12	2009	4	Andhra Pradesh	1	Adilabad	ST	Kotnak Rames
15	13	2009	4	Andhra Pradesh	1	Adilabad	ST	Mesram Nago
16	14	2009	4	Andhra Pradesh	1	Adilabad	ST	Ade Tukaram

Key observations about the dataset:

1. The data set has loksabha general elections/byelections candidate level details with 26,114 rows and 19 columns.
2. It has missing data which is a blank data's in column month, candidate_type, sex column, party, vote share percentage, margin, margin_percentage
3. It has inconsistent data in state-name, pc_name and candidate name, sex, party, and duplicate data's
4. It has no duplicates

Handling Blank datas:

Cleaning for Column Month

The column month contains blank value. I have checked this by using the filter option in header. Since the column has only month 4 which is April, I am replacing the blank cells to 4(April) as well.

Formula Used – Ctrl + H for replace all

C2 ✖ ✓ fx 4

	A	B	C	D	E	F
1	id	year	month	state_name	pc_code	pc_name
2	0	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Isl
3	1	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Isl
4	2	2009	4	An		Isl
5	3	2009	4	An		Isl
6	4	2009	4	An		Isl
7	5	2009	4	An		Isl
8	6	2009	4	An		Isl
9	7	2009	4	An		Isl
10	8	2009	4	An		Isl
11	9	2009	4	An		Isl
12	10	2009	4	An		Isl
13	11	2009	4	Andhra Pradesh	1	Adilabad
14	12	2009	4	Andhra Pradesh	1	Adilabad
15	13	2009	4	Andhra Pradesh	1	Adilabad

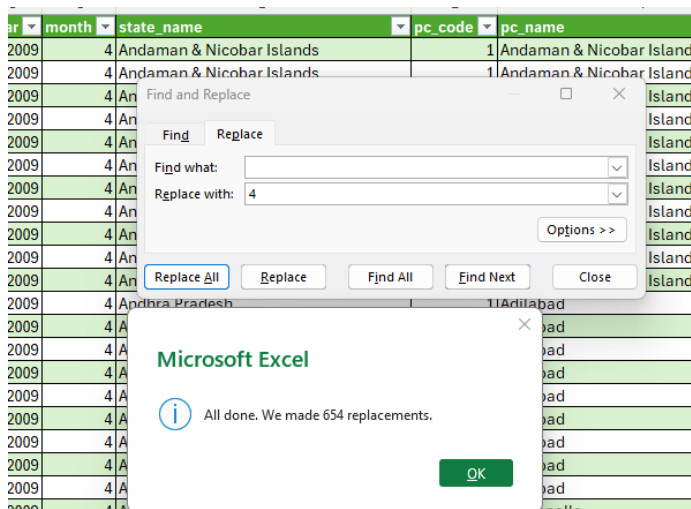
Find and Replace

Find what:

Replace with: 4

Options >>

Replace All Replace Find All Find Next Close



Cleaning for Candidate_type column:

The candidate_type column has blank values. Firstly, I am going to compare the column pc_type and candidate type since both columns have some similar values by using the below formula.

Formula used - =IF([@pc_type]=[@candidate_type],"Match","Unmatched")

Get & Transform Data

Queries & Connections

Sort & Filter

Data Tools

46

<

Where it has matched entries of 17,195 and unmatched entries of 8,921(includes blanks of 6,510) actual non-blank mismatches: 2,411, therefore I am keeping both the columns as it is and just going to fill the blank data of candidate_type with the pc_type values using below formula

Formula Used -

=IF(LEN(TRIM([@candidate_type]))=0,[@pc_type],[@candidate_type])

Get & Transform Data				Queries & Connections		Sort & Filter		Data Tools		Forecast		Outline	
J2				=IF(LEN(TRIM([@candidate_type]))=0,[@pc_type],[@candidate_type])									
B		C		D		E		F		G		H	
1	year	month	state_name	pc_code	pc_name	pc_type	candidate_name		candidate_type	Column1			
2	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Bishnu Pada Ray		GEN	GEN			
3	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Kuldeep Rai Sharma		GEN	GEN			
4	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Tapan Kumar Bepari		GEN	GEN			
5	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. P. R. Ganesan		GEN	GEN			
6	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Smti. R. S. Uma Bharathy		GEN	GEN			
7	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Pradeep Kumar Ekka		GEN	GEN			
8	2009	4	Andaman & Nicobar Islands	1	Andaman & Nicobar Islands	GEN	Shri. Vakiath Valannil Khalid		GEN	GEN			

Cleaning for Sex Column:

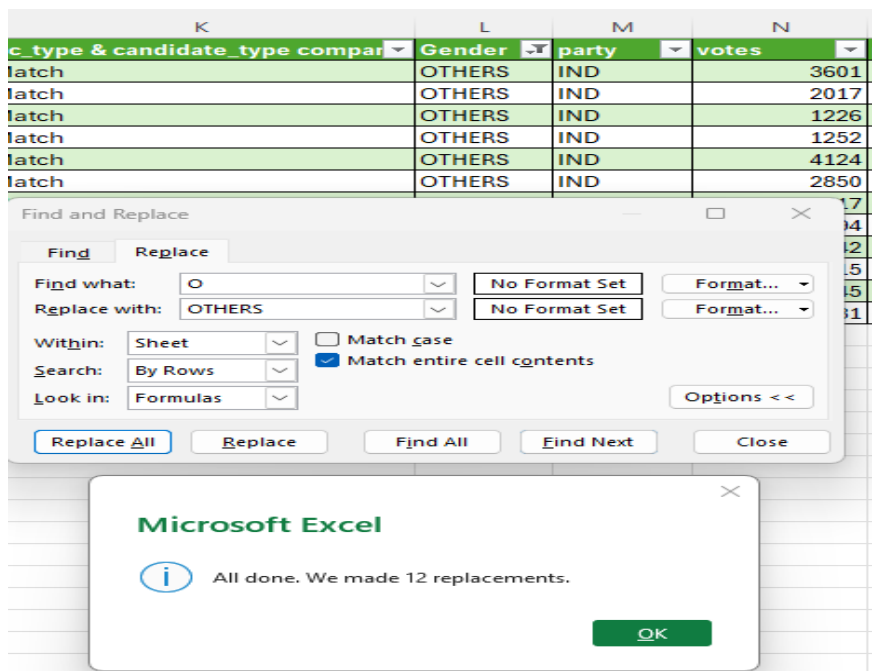
I changed the column header name to Gender as it is more appropriate. It has inconsistent data.

I selected the Gender column and replaced the below by replace all function – Ctrl + H

F -> FEMALE

J	K	L	M	N	O
idate_type	pc_type & candidate_type compar	Gender	party	votes	position
	Match	M	BJP	75211	
	Match	M	INC	72221	
	Match	M	CPM	7190	
	Match	M	RJD	4916	
	Match	F	NCP	4696	
	Match	M	JDP	1785	
	Match	M	IND	1480	
	Match	M	BSP	789	
	Match	M	CPI(ML)(L)	734	
	Match	M	IND	683	

Replaced NOTA entry to “Unknown” and O to Others



Cleaning Party Column:

The party column has blank cells. However, those blank cells have the candidate's name as Nota, therefore filling the blank ones as nota

	H	I	J	K	L	M	N
ve	candidate_name	candidate_type	candidate_type	pc_type & candidate_type compar	Gender	party	vote
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		ST	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		ST	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		SC	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		SC	Unmatched	UNKNOWN	NOTA	
	Nota		ST	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	
	Nota		GEN	Unmatched	UNKNOWN	NOTA	

Standardizing Formats(Inconsistencies)

I am going to clean the below columns as these has extra spaces and some inconsistencies in the text

- state_name
- pc_name
- candidate_name
- party

Excel Formula Used:

=TRIM(CLEAN(A2))

=PROPER(TRIM([@[pc_name]]))

=UPPER(TRIM([@party]))

This removes:

- Leading spaces
- Trailing spaces
- Hidden characters
- Changing cases

State name before Trim:

Get & Transform Data

Queries & Connections

Sort & Filter

D2

:

=TRIM(CLEAN([@[state_name]]))

	A	B	C	D	E	F
1	id	year	month	state_name	Column1	pc_code
2	0	2009	4	Andaman & Nicobar Islands	=TRIM(CLEAN([@[state_name]]))	
3	1	2009	4	Andaman & Nicobar Islands		
4	2	2009	4	Andaman & Nicobar Islands		
5	3	2009	4	Andaman & Nicobar Islands		
6	4	2009	4	Andaman & Nicobar Islands		
7	5	2009	4	Andaman & Nicobar Islands		
8	6	2009	4	Andaman & Nicobar Islands		
9	7	2009	4	Andaman & Nicobar Islands		

State name after Trim:

Get & Transform Data

Queries & Connections

Sort & Filter

D1

✕

✓

fx

state_name

	A	B	C	D	E	
1	id	year	month	state_name	Column1	pc_c
2	0	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
3	1	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
4	2	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
5	3	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
6	4	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
7	5	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
8	6	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
9	7	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
10	8	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
11	9	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	
12	10	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	

PC_name before Proper:

	A	B	C	D	E	F	G	H
1	id	year	month	state_name	state_name1	pc_code	pc_name	pc_name1
2	0	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicoba	=TRIM(CLEAN([@[pc_name]]))
3	1	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
4	2	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
5	3	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
6	4	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
7	5	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
8	6	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	
9	7	2009	4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	

PC-name after Trim and Proper:

Get & Transform Data

Queries & Connections

Sort & Filter

Data Tools

H3

=TRIM(CLEAN([@[pc_name]]))

	A	B	C	D	E	F	G	H
1	id	year	month	state_name	state_name1	pc_code	pc_name	pc_name1
2		0	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
3		1	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
4		2	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
5		3	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
6		4	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
7		5	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
8		6	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands
9		7	2009	4 Andaman & Nicobar Islands	Andaman & Nicobar Is		1 Andaman & Nicoba	Andaman & Nicobar Islands

=PROPER(TRIM([@[pc_name]]))						
C	D	E	F	G	H	
th	state_name	State Name	Constituency Cod	pc_name	Constituency Name	Constituency
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isl	GEN

Candidate name before TRIM:

	I	J	K	L	M
	pc_type	candidate_name	candidate_name1	candidate	candidate
	GEN	Shri. Bishnu Pada Ray	=TRIM(CLEAN([@[candidate_name]]))		GEN
	GEN	Shri. Kuldeep Rai Sharma		GEN	GEN
	GEN	Shri Tapan Kumar Bepari		GEN	GEN
	GEN	Shri. P. R. Ganeshan		GEN	GEN
	GEN	Smti. R. S. Uma Bharatthy		GEN	GEN
	GEN	Shri. Pradeep Kumar Ekka		GEN	GEN
	GEN	Shri. Vakiath Valappil Khalid		GEN	GEN
	GEN	Shri. M. S. Mohan		GEN	GEN
	GEN	Shri. N. K. P. Nair		GEN	GEN
	GEN	Dr. Thankachan		GEN	GEN

Font

Alignment

Number

fx

=PROPER(TRIM([@[candidate_name]]))

I	J	K	L	M
type	candidate_name	candidate_name1	candidate_t	candidate_type
	Shri. Bishnu Pada Ray	=PROPER(TRIM([@[candidate_name]]))		
	Shri. Kuldeep Rai Sharma		GEN	GEN
	Shri Tapan Kumar Bepari		GEN	GEN
	Shri. P. R. Ganeshan		GEN	GEN
	Smti. R. S. Uma Bharatthy		GEN	GEN
	Shri. Pradeep Kumar Ekka		GEN	GEN
	Shri. Vakiath Valappil Khalid		GEN	GEN
	Shri. M. S. Mahan		GEN	GEN

Candidate name after TRIM & PROPER:

Font		Alignment		Number		Styles		Cells	
=PROPER(TRIM([@[candidate_name]]))									
C	D	E	F	G	H	I	J	K	
nth	state_name	State Name	Constituency Code	pc_name	Constituency Name	Constituency Type	candidate_name	Candidate Name	car
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Shri. Bishnu Pada Ray	Shri. Bishnu Pada Ra	GE
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Shri. Kuldeep Rai Sharm	Shri. Kuldeep Rai Sha	GE
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Shri Tapan Kumar Bepar	Shri Tapan Kumar Be	GE
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Shri. P. R. Ganeshan	Shri. P. R. Ganeshan	GE
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Smti. R. S. Uma Bharatth	Smti. R. S. Uma Bhar	GE
4	Andaman & Nicobar Islands	Andaman & Nicobar Is	1	Andaman & Nicobar Islands	Andaman & Nicobar Isla	GEN	Shri. Pradeep Kumar Ek	Shri. Pradeep Kumar	GE

Party Code before Trim:

	O	P	Q	
andidate_type	Gender	party	Party Code	Candid
	MALE	BJP	=UPPER(TRIM([@[party]]))	
	MALE	INC	INC	
	MALE	CPM	CPM	
	MALE	RJD	RJD	

Party Code after Trim:

fx =UPPER(TRIM([@[party]]))						
	L	M	N	O	P	Q
andidate_type	Candidate Category	pc_type & candidate_type	Gender	party	Party Code	C
IN	GEN	Match	MALE	BJP	BJP	
IN	GEN	Match	MALE	INC	INC	
IN	GEN	Match	MALE	CPM	CPM	
IN	GEN	Match	MALE	RJD	RJD	
IN	GEN	Match	FEMALE	NCP	NCP	
IN	GEN	Match	MALE	JDP	JDP	
IN	GEN	Match	MALE	IND	IND	
IN	GEN	Match	MALE	BSP	BSP	
IN	GEN	Match	MALE	CPI(ML)(L)	CPI(ML)(L)	

Manually Changing some Party Codes:

Left([@party]))

	N	O	P	Q	R	
y	pc_type & candidate_type	Gender	party	Party Code	Candidate Vo	Candidate
	Match	male	Aihra National Party	Aihra National Party	1279	

	O	P	Q	R	
type	Gender	party	Party Code	Candidate Vo	Candid
	male	Aihra National Party	ANP	1279	

Manually Formatted the column headers names for easy reference

Q	R	S	T	U	V	W	X	
Party Code	Candidate Votes	Candidate Position/Rank	Total Valid Votes	Registered_Voters	Turnout %	Vote Share %	Victory Margin	Victory
BJP	75211	1	170103	265108	64.16	44.21	2990	
INC	72221	2	170103	265108	64.16	42.46	65031	

Creating Calculated Fields:

Election Category:

I have created a new column named “Election Category” to show whether the election is for general election or by election using the if else excel function

Formula used : =IF(OR([@Year]=2009,[@Year]=2014,[@Year]=2019),"General Election","By-Election")

Get & Transform Data						Queries & Connections		Sort & Filter	
Data Text/CSV Web Range Sources Connections						All Workbook Links		Sort & Filter	
D2						=IF(OR([@Year]=2009,[@Year]=2014,[@Year]=2019),"General Election","By-Election")			
1	A	B	C	D	E	F			
1	Id	Year	Month	Election Category	state_name	State Name			
2	0	2009	4	=IF(OR([@Year]=2009,[@Year]=2014,[@Year]=2019),"General Election","By-Election")					
3	1	2009	4		Andaman & Nicobar Islands	Andaman			
4	2	2009	4		Andaman & Nicobar Islands	Andaman			
5	3	2009	4		Andaman & Nicobar Islands	Andaman			
6	4	2009	4		Andaman & Nicobar Islands	Andaman			
7	5	2009	4		Andaman & Nicobar Islands	Andaman			

Data ▾ Text/CSV Web Range Sources Connections All ▾ Workbook Links Get & Transform Data Queries & Connections Sort & Filter

D3 :

	A	B	C	D	E	F
	Id	Year	Month	Election Category	state_name	State Name pc
1	0	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
2	1	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
3	2	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
4	3	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
5	4	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
6	5	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
7	6	2009	4	General Election	Andaman & Nicobar Islands	Andaman &
8	7	2009	4	General Election	Andaman & Nicobar Islands	Andaman &

Election Result:

I have created a new column named “Election Result” to show the status win or lost by using the candidate position/rank column using the if else excel function

Formula used : =IF([@[Candidate Position/Rank]]=1,"Winner","Lost")

	S	T	U	V
es ▾	Candidate Position ▾	Election Result ▾	Total Valid Votes ▾	Registered_Vot ▾
75211	1	=IF([@[Candidate Position/Rank]]=1,"Winner","Lost")		
72221	2		170103	265108
7190	3		170103	265108
4916	4		170103	265108
4696	5		170103	265108

R	S	T	U	
Candidate Votes ▾	Candidate Position ▾	Election Result ▾	Total Valid Votes ▾	Reg
75211	1	Winner	170103	
72221	2	Lost	170103	
7190	3	Lost	170103	
4916	4	Lost	170103	
4696	5	Lost	170103	
1785	6	Lost	170103	
1480	7	Lost	170103	
789	8	Lost	170103	

Turnout Category:

I have created a new column named “Turnout Category” to show which category it falls by using the turnout % column using the if else excel function

Formula used : =IF([@[Turnout %]]>=70,"High",IF([@[Turnout %]]>=50,"Medium","Low"))

	W	X	Y	Z	AA
Vot	Turnout %	Turnout Category	Vote St	Victory Margin	Victory
265108	64.16	=IF([@[Turnout %]]>=70,"High",IF([@[Turnout %]]>=50,"Medium","Low"))			
265108	64.16				
265108	64.16		4.23	2274	1.34
265108	64.16		2.89	220	0.13
265108	64.16		2.76	2911	1.71

	W	X	Y	Z
t	Turnout %	Turnout Category	Vote St	Victory Margin
108	64.16	Medium	44.21	2
108	64.16	Medium	42.46	65
108	64.16	Medium	4.23	2
108	64.16	Medium	2.89	
108	64.16	Medium	2.76	2
108	64.16	Medium	1.05	
108	64.16	Medium	0.87	
108	64.16	Medium	0.46	
108	64.16	Medium	0.43	
108	64.16	Medium	0.4	
108	64.16	Medium	0.23	
211	76.34	High	43.11	115
211	76.34	High	29.78	144

Victory Margin Category:

I have created a new column named "Victory Margin Category" to show which category it falls by using the victory margin % column using the if else excel function

Formula used : =IF([@[Victory Margin %]]<5,"Very Close",IF([@[Victory Margin %]]<10,"Close",IF([@[Victory Margin %]]<20,"Moderate Lead","Large Lead")))

	AA	AB	AC	AD	A
gin	Victory Margin %	Victory Margin C			
2990	1.76	=IF([@[Victory Margin %]]<5,"Very Close",IF([@[Victory Margin %]]<10,"Close",IF([@[Victory Margin %]]<20,"Moderate Lead","Large Lead")))			
65031	38.23				
2274	1.34				
220	0.13				
2911	1.71				
305	0.18				
604	0.44				

Z	AA	AB	AC
Margin	Victory Margin %	Victory Margin C	
2990	1.76	Very Close	
65031	38.23	Large Lead	
2274	1.34	Very Close	
220	0.13	Very Close	
2911	1.71	Very Close	
205	0.18	Very Close	

Candidate vs Constituency Type

I have created a new column named “Candidate vs Constituency Type” to append the constitution type and candidate category by using the Concat excel function

Formula used : =CONCAT([@[Constituency Type]] & "-" & [@[Candidate Category]])

	M	N	O	P	
type	Candidate Category	pc_type & candidate_type	Candidate vs Constituency	Gender	party
	GEN	Match	=CONCAT([@[Constituency Type]] & "-" & [@[Candidate Category]])		
	GEN	Match		MALE	INC
	GEN	Match		MALE	CPM
	GEN	Match		MALE	RJD
	GEN	Match		FEMALE	NCP
	GEN	Match		MALE	JDP

	M	N	O	P
e_type	Candidate Category	pc_type & candidate_type	Candidate vs Constituency	Gender
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	FEMALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE
	GEN	Match	GEN-GEN	MALE

Creating Dimension Tables:

Dim_Constituency:

I have created a dimension table named constituency by following the below steps.

Steps

1. Created a new sheet called: Dim_Constituency
2. Copied constituency name, state name and constituency type from main dataset

3. Pasted into Dim_Constituency sheet
4. Selected all data.
5. Go to: Data → Remove Duplicates
6. Check in all the three columns
7. Click OK.

Clipboard Font Alignment			
A1	✕ ✓ fx	Constituency Name	
	A	B	C
1	Constituency Name	State Name	Constituency Type
2	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
3	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
5	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
6	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
7	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
8	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
9	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
10	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
11	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
12	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
13	Adilabad	Andhra Pradesh	ST
14	Adilabad	Andhra Pradesh	ST
15	Adilabad	Andhra Pradesh	ST
16	Adilabad	Andhra Pradesh	ST

1	Constituency Name	State Name	Constituency Type
2	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
3	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
4	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
5	Andaman &		
6	Andaman &		
7	Andaman &		
8	Andaman &		
9	Andaman &		
10	Andaman &		
11	Andaman &		
12	Andaman &		
13	Adilabad		
14	Adilabad		
15	Adilabad		
16	Adilabad		
17	Adilabad		
18	Adilabad		
19	Adilabad	Andhra Pradesh	ST
20	Adilabad	Andhra Pradesh	ST
21	Adilabad	Andhra Pradesh	ST
22	Peddapalle	Andhra Pradesh	SC

Remove Duplicates

?

✕

To delete duplicate values, select one or more columns that contain duplicates.

Select All

Unselect All

☒ My data has headers

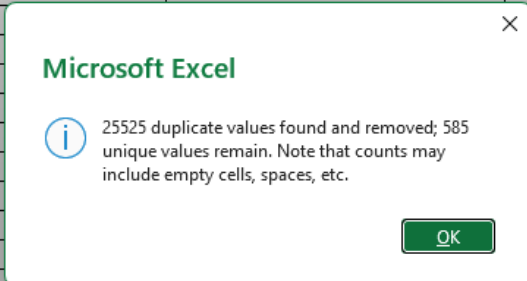
Columns

☒ Constituency Name
☒ State Name
☒ Constituency Type

OK

Cancel

	A	B	C
1	Constituency Name	State Name	Constituency Type
2	Andaman & Nicobar Islands	Andaman & Nicobar Islands	GEN
3	Adilabad	Andhra Pradesh	ST
4	Peddapalle	Andhra Pradesh	SC
5	Karimnagar	Andhra Pradesh	GEN
6	Nizamabad	Andhra Pradesh	GEN
7	Zahirabad	Andhra Pradesh	GEN
8	Medak	Andhra Pradesh	GEN
9	Malkajgiri		N
10	Secundrabad		N
11	Hyderabad		N
12	Chelvella		N
13	Mahbubnagar		N
14	Nagarkurnool		N
15	Nalgonda		N
16	Bhongir		N
17	Warangal		
18	Mahabubabad		
19	Khammam	Andhra Pradesh	GEN
20	Aruku	Andhra Pradesh	ST
21			GEN



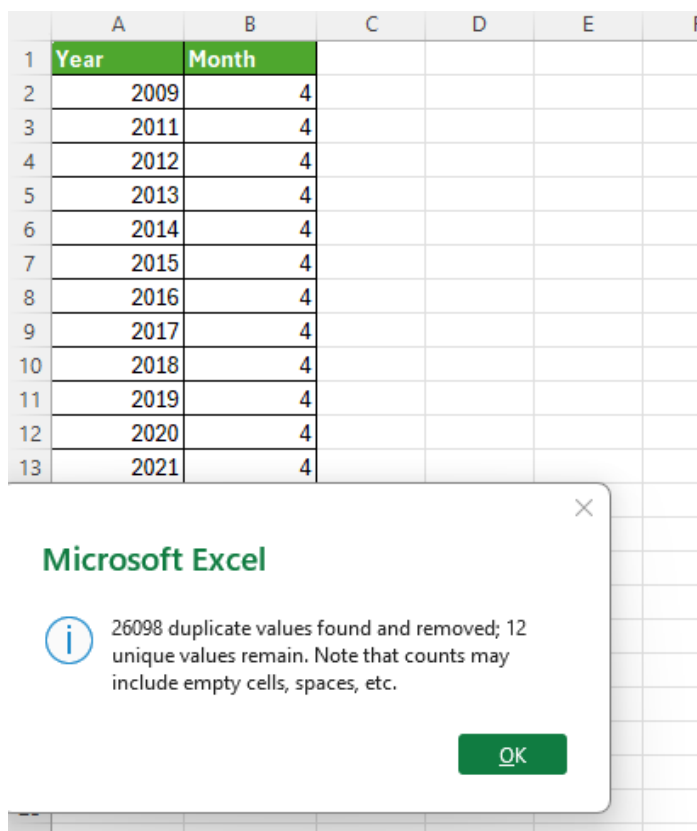
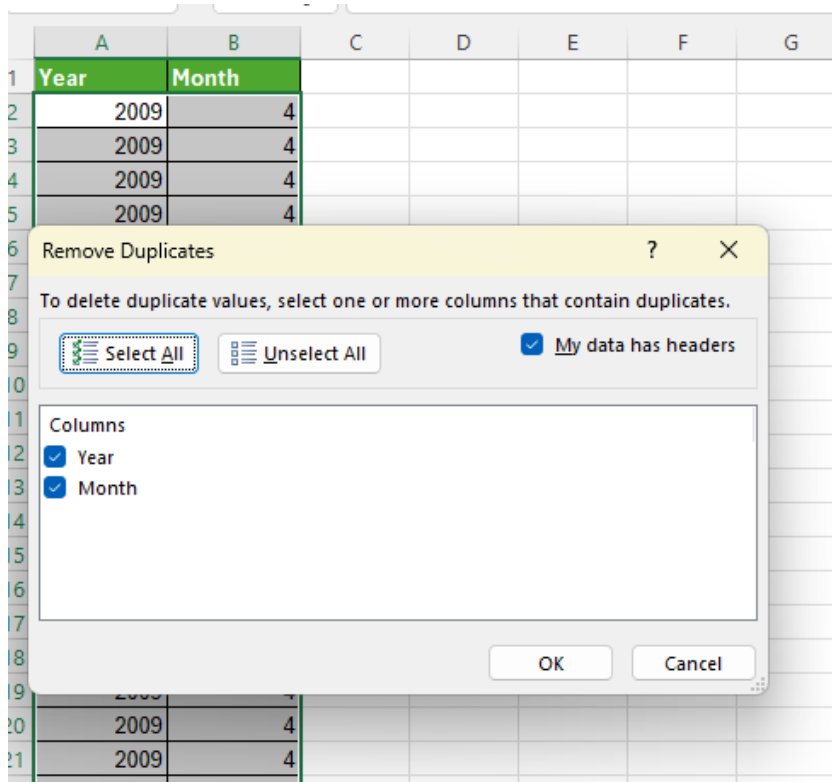
Creating Dim_Year Table:

I have created a dimension table named Year by following the below steps.

Steps

1. Created a new sheet called: Dim_Year
2. Copied year and month from main dataset
3. Pasted into Dim_Year sheet
4. Selected all data.
5. Go to: Data → Remove Duplicates
6. Check in all the three columns
7. Click OK

Get & Transform Data					
A1					
	A	B	C	D	E
1	Year	Month			
2	2009	4			
3	2009	4			
4	2009	4			
5	2009	4			
6	2009	4			
7	2009	4			
8	2009	4			
9	2009	4			
10	2009	4			
11	2009	4			
12	2009	4			



Then, added election_cycle column by using helper column

Clipboard				Font			
C2				=A2 & " Election"			
	A	B	C	D			
1	Year	Month	Election_Cycle				
2	2009	4	2009 Election				
3	2011	4	2011 Election				
4	2012	4	2012 Election				
5	2013	4	2013 Election				
6	2014	4	2014 Election				
7	2015	4	2015 Election				
8	2016	4	2016 Election				
9	2017	4	2017 Election				
10	2018	4	2018 Election				
11	2019	4	2019 Election				
12	2020	4	2020 Election				
13	2021	4	2021 Election				

Creating Dim_Candidate Table:

I have created a dimension table named Candidate by following the below steps.

Steps

1. Created a new sheet called: Dim_Candidate
2. Copied candidate name column from main dataset
3. Pasted into Dim_Candidate sheet
4. Selected only the candidate name column
5. Go to: Data → Remove Duplicates
6. Click OK
7. Added a columns gender and candidate category by using Xlookup
8. Formula Used - =XLOOKUP(A2,'loksabha-general-elections'!\$K:\$K,'loksabha-general-elections'!\$P:\$P,"Not Found")
9. Formula Used - =XLOOKUP(A2,'loksabha-general-elections'!\$K:\$K,'loksabha-general-elections'!\$M:\$M,"Not found")

	A	B	C	D
1	Candidate Name			
2	Shri. Bishnu Pada Ray			
3	Shri. Kuldeep Rai Sharma			
4	Shri Tapan Kumar Bepari			
5	Shri. P. R. Ganeshan			
6	Smti. R. S. Uma Bharatthy			
7	Shri. Pradeep Kumar Ekka			
8	Shri. Vakiath Valappil Khalid			
9	Shri. M. S. Mohan			
20	Nethavat Ramdas			
21	Athram Laxman Rao			

Microsoft Excel



4125 duplicate values found and removed; 21985 unique values remain. Note that counts may include empty cells, spaces, etc.

OK

Clipboard		Font		Alignment		Number	
B2							
=XLOOKUP(A2,'loksabha-general-elections'!\$K:\$K,'loksabha-general-elections'!\$P:\$P,"Not Found")							
A	B	C	D	E	F	G	H
Candidate Name	Gender						
Shri. Bishnu Pada Ray	MALE						
Shri. Kuldeep Rai Sharma							
Shri Tapan Kumar Bepari							

B2							
=XLOOKUP(A2,'loksabha-general-election							
A	B	C	D	E			
Candidate Name	Gender						
Shri. Bishnu Pada Ray	MALE						
Shri. Kuldeep Rai Sharma	MALE						
Shri Tapan Kumar Bepari	MALE						
Shri. P. R. Ganeshan	MALE						
Smti. R. S. Uma Bharatthy	FEMALE						
Shri. Pradeep Kumar Ekka	MALE						
Shri. Vakiath Valappil Khalid	MALE						
Shri. M. S. Mohan	MALE						
Shri. N. K. P. Nair	MALE						
Dr. Thankachan	MALE						

Clipboard		Font		Alignment		Number	
C2							
=XLOOKUP(A2,'loksabha-general-elections'!\$K:\$K,'loksabha-general-elections'!\$M:\$M,"Not found")							
A	B	C	D	E	F	G	H
Candidate Name	Gender	Candidate Category					
Shri. Bishnu Pada Ray	MALE	GEN					
Shri. Kuldeep Rai Sharma	MALE						
Shri Tapan Kumar Bepari	MALE						
Shri. P. R. Ganeshan	MALE						
Smti. R. S. Uma Bharatthy	FEMALE						
Shri. Pradeep Kumar Ekka	MALE						
Shri. Vakiath Valappil Khalid	MALE						
Shri. M. S. Mohan	MALE						

A	B	C	D
Candidate Name	Gender	Candidate Category	
Shri. Bishnu Pada Ray	MALE	GEN	
Shri. Kuldeep Rai Sharma	MALE	GEN	
Shri Tapan Kumar Bepari	MALE	GEN	
Shri. P. R. Ganeshan	MALE	GEN	
Smti. R. S. Uma Bharatthy	FEMALE	GEN	
Shri. Pradeep Kumar Ekka	MALE	GEN	
Shri. Vakiath Valappil Khalid	MALE	GEN	
Shri. M. S. Mohan	MALE	GEN	
Shri. N. K. P. Nair	MALE	GEN	
Dr. Thankachan	MALE	GEN	
Shri. T. Ali	MALE	GEN	

Data Modelling:

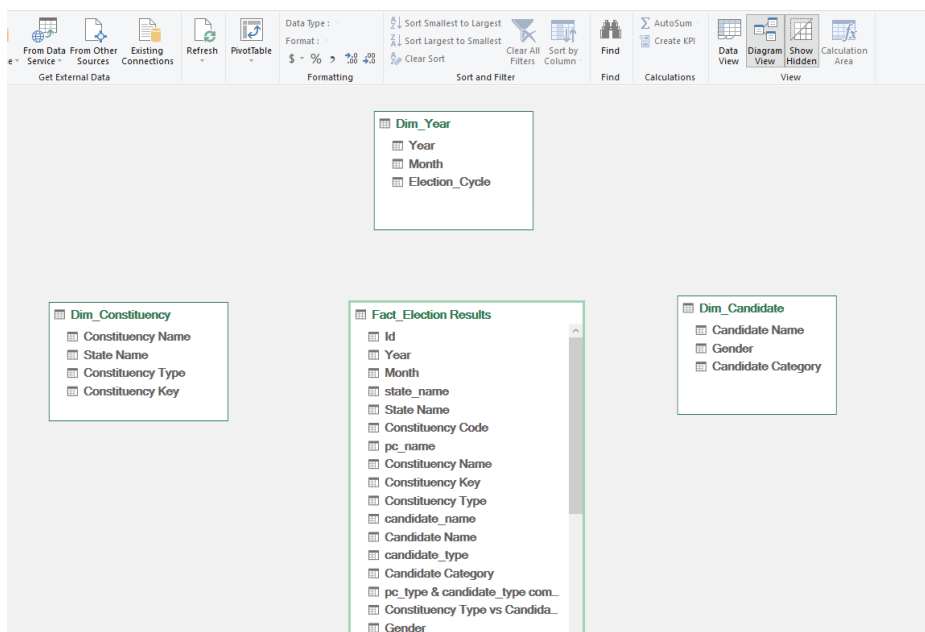
Step 1: Ensure All Tables Are Excel Tables by clicking every tables to Ctrl+T by clicking my table has headers.

Step 2: Then, select table -> Goto Power Pivot tab -> Click Add to data model

Step 3: Do the above step for all created tables to load the tables to data model

Step 4: In view tab -> Click Diagram View to check their relations

Step 5: Create relationships by dragging the matching columns from main fact table to the created dimension tables.



Final Star Schema:

